

◆ PROJECT MASTER PLAN ◆

SkillForge

AI-Powered Developer Growth Platform

The app that tells you exactly what's missing between YOU and your dream job.

Project	SkillForge — AI Developer Growth Platform
Author	Syed Humaam Ahmed (Kairo)
Build Plan	Kimi 2.5/2.6 (Frontend) + GLM 5.1 (Backend)
Fallback	Google Antigravity (Full Build)
Version	v1.0 2026
Status	Planning Phase — Ready to Build

React JS

Node.js

MongoDB

GitHub OAuth

Gemini AI

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01 — Project Overview & Vision

SkillForge is an AI-powered developer growth platform designed to bridge the gap between where a developer is today and where they want to be. Unlike generic learning platforms, SkillForge is deeply personalized — it reads your GitHub profile, analyzes real job listings, identifies your exact skill gaps, and builds a custom roadmap just for you. This is a product that solves a real problem every developer faces.

Goal	Description
Real Problem Solver	Answers 'what should I learn next?' with data, not guesses.
GitHub Integration	Analyzes real repos, languages, commit activity, and tech usage.
AI Skill Gap Engine	Compares your profile against job listings using LLM analysis.
Personalized Roadmaps	30/60/90 day plans tailored to YOUR specific gaps.
Recruiter-Ready Profile	Public shareable page showing your growth journey.
Resume AI Generator	Turns GitHub activity into polished resume bullet points.
Portfolio Showpiece	Demonstrates full-stack + AI + OAuth — a complete product.

02 — Problem Statement

Most developers are stuck in a cycle: watch a random tutorial, build a project, apply for jobs, get rejected, repeat. The root cause is they never identify their specific skill gaps relative to the job they actually want. SkillForge solves this with data-driven, AI-powered gap analysis.

Developer Pain Point	SkillForge Solution
'I don't know what to learn next'	AI analyzes your GitHub + target job and tells you exactly
Random tutorial consumption	Structured 30/60/90 day personalized roadmap
No way to measure progress	Daily challenges + progress dashboard + skill radar chart
Resume bullet points are weak	AI generates resume bullets directly from GitHub activity
No proof of growth for recruiters	Public shareable profile page showing journey and stats
Generic interview prep	Mock interview questions tailored to YOUR specific gaps

03 — Tech Stack — Full Breakdown

Frontend

Library / Tool	Version	Purpose
React JS	v18+	Core UI framework — component-based architecture
Vite	v5	Lightning fast build tool and dev server
Tailwind CSS	v3	Utility-first CSS — rapid UI development
React Router	v6	Client-side routing between all pages
Zustand	Latest	Lightweight global state management
Recharts	Latest	Beautiful charts — radar, bar, line, progress
Framer Motion	Latest	Smooth page transitions and micro-animations
Lucide React	Latest	Clean icon library
React Query	v5	Server state management and API caching
Axios	Latest	HTTP client for backend API calls

Backend

Library / Tool	Version	Purpose
Node.js	v20 LTS	Backend JavaScript runtime
Express.js	v4	REST API framework — fast and minimal
Mongoose	v8	MongoDB ODM — schema modeling and queries
Passport.js	Latest	GitHub OAuth 2.0 authentication strategy
express-session	Latest	Session management for authenticated users
connect-mongo	Latest	Store sessions in MongoDB
jsonwebtoken	Latest	JWT generation for API auth
bcryptjs	Latest	Password hashing utility
node-cron	Latest	Daily challenge scheduler
dotenv	Latest	Environment variable management
cors	Latest	Cross-origin resource sharing middleware
helmet	Latest	HTTP security headers

Database

Library / Tool	Version	Purpose
MongoDB Atlas	Free Tier	Cloud NoSQL database — flexible document model
Mongoose ODM	v8	Schema definitions, validation, and queries

AI Layer

Library / Tool	Version	Purpose
Google Gemini API	2.0 Flash	Skill gap analysis + roadmap generation
OpenAI / Claude / Groq	BYOK	Optional — user provides own key

Third-Party APIs

Library / Tool	Version	Purpose
GitHub REST API	v3	Fetch repos, languages, commits, user stats
GitHub OAuth 2.0	—	Login with GitHub — no password needed

DevOps / Deploy

Library / Tool	Version	Purpose
GitHub	—	Source control + release management
Vercel	—	Frontend hosting — auto-deploys from GitHub
Railway / Render	Free Tier	Node.js + Express backend hosting
MongoDB Atlas	Free Tier	Cloud database — no server needed

04 — Core Features — Detailed

GitHub Profile Analyzer

- User logs in with GitHub OAuth — one click, no password.
- Fetches all public repos, programming languages, commit frequency, and stars.
- Detects tech stack from repo contents and language percentages.
- Generates a 'Current Skill Level' radar chart across 8 skill categories.
- Shows GitHub contribution heatmap and activity timeline.
- Identifies most active time periods and productivity patterns.

Job Listing Analyzer

- User pastes any job listing URL or raw job description text.
- AI extracts required skills, tools, experience level, and keywords.
- Generates a 'Job Requirements' skill breakdown with priority levels.
- Compares job requirements against user's GitHub profile automatically.
- Calculates a 'Match Score' (0-100%) for the target role.
- Highlights top 5 most critical missing skills to focus on.

Skill Gap Report

- Visual side-by-side radar chart: Your Skills vs Job Requirements.
- Colour-coded gap indicators: Red (critical), Amber (moderate), Green (strong).
- Detailed breakdown per skill: current level, required level, gap size.
- Prioritized list of skills to learn ordered by impact on match score.
- Downloadable PDF report of the full skill gap analysis.
- Re-analyzable — update as you learn and watch match score climb.

Personalized Roadmap Generator

- AI generates a 30, 60, or 90-day learning roadmap based on skill gaps.
- Each day has a specific task: read, watch, build, or practice.
- Resources included: best YouTube videos, docs, and free courses per skill.
- Roadmap broken into weekly milestones with progress checkboxes.
- User can mark tasks complete — progress tracked with a progress bar.
- Roadmap regenerates automatically when new gaps are detected.

Daily Challenge System

- Every day at midnight a new coding challenge is assigned to the user.
- Challenges are calibrated to the user's current skill gaps (not generic).
- Challenge types: coding problem, mini-project, concept explanation, or quiz.
- User submits their solution or marks challenge as complete.
- Streak tracker: consecutive days completed shown with flame icon.
- Monthly challenge history with performance summary.

AI Mentor Chat

- Persistent chat interface powered by Gemini AI (user's BYOK key).
- AI has full context: user's GitHub stats, skill gaps, and roadmap.
- Can answer questions like 'explain my biggest weakness' or 'what to build next'.
- Streaming responses for real-time conversation feel.
- Chat history saved per session in localStorage.
- Pre-built prompt buttons: 'Explain my gap', 'Suggest a project', 'Mock interview'.

Resume Bullet Generator

- Analyzes GitHub repositories and commit messages.
- Generates professional resume bullet points per project.
- Format: Action verb + technology + measurable outcome.
- Example: 'Built a RESTful API with Node.js handling 500+ daily requests'.
- User can edit, regenerate, or copy individual bullet points.
- Export all bullets as a formatted .txt or .pdf file.

Public Profile Page

- Shareable URL: skillforge.app/u/kairo
- Displays GitHub stats, current skills radar, active roadmap progress.
- Shows streak count, challenges completed, and match score history.
- Recruiters can view profile without logging in.
- Toggle public/private per section for privacy control.
- Copy profile link button + social share preview card (og:image).

05 — Additional Features

Feature	Description	Priority
Dark / Light Mode	Full theme toggle — preference saved in localStorage	HIGH
Mock Interview Mode	AI asks role-specific interview questions + evaluates answers	HIGH
Skill Progress Timeline	Visual timeline showing skill growth over weeks/months	HIGH
Roadmap PDF Export	Download full personalized roadmap as a formatted PDF	HIGH
Notification System	Daily reminder: 'Your challenge for today is ready!'	MEDIUM
Leaderboard	Compare streak and challenge scores with other SkillForge users	MEDIUM
Project Suggestions	AI suggests portfolio projects that fill your skill gaps	HIGH
GitHub Repo Deep Scan	Analyze code quality, patterns, and tech depth per repo	MEDIUM
Keyboard Shortcuts	Power user shortcuts for all major actions	LOW
Multi-language UI	English / Urdu language toggle	LOW

06 — All Pages & Screens

Screen Name	Route	Description
Landing Page	/	Hero + features + CTA 'Login with GitHub' — public
GitHub OAuth Callback	/auth/callback	Processes GitHub OAuth token — redirects to dashboard
Dashboard	/dashboard	Welcome back card + GitHub stats + daily challenge card + quick actions
GitHub Analyzer	/analyze	Fetches and displays full GitHub profile analysis + skill radar chart
Job Analyzer	/job	Paste job URL or text — AI extracts requirements + match score
Skill Gap Report	/gap-report	Visual radar comparison + prioritized gap list + download PDF
My Roadmap	/roadmap	Active 30/60/90 day roadmap with checkboxes + progress bar
Daily Challenges	/challenges	Today's challenge + streak tracker + challenge history
AI Mentor Chat	/mentor	Full-screen chat interface with AI mentor context
Resume Generator	/resume	GitHub repo selector + AI bullet generator + export options
Public Profile	/u/:username	Public shareable page — recruiter-friendly view
Settings	/settings	API key, theme, notifications, profile privacy controls
Leaderboard	/leaderboard	Community challenge streaks and scores
Mock Interview	/interview	AI interviewer session — role-specific Q&A + evaluation
404 Not Found	*	Branded friendly error page

07 — Folder Structure

```
skillforge/
  ■■■ client/ ← React Frontend (Kimi builds this)
  ■ ■■■ public/
  ■ ■■■ index.html
  ■ ■■■ src/
  ■ ■■■ components/
  ■ ■ ■■■ Layout/ ← Navbar, Sidebar, Footer
  ■ ■ ■■■ Dashboard/ ← Stats cards, welcome section
  ■ ■ ■■■ GitHubAnalyzer/ ← Repo cards, radar chart, heatmap
  ■ ■ ■■■ JobAnalyzer/ ← Job paste input, match score
  ■ ■ ■■■ SkillGapReport/ ← Side-by-side radar, gap list
  ■ ■ ■■■ Roadmap/ ← Timeline, checkboxes, progress
  ■ ■ ■■■ Challenges/ ← Challenge card, streak tracker
  ■ ■ ■■■ MentorChat/ ← Streaming chat UI
  ■ ■ ■■■ ResumeGen/ ← Repo selector, bullet cards
  ■ ■ ■■■ PublicProfile/ ← Recruiter view components
  ■ ■ ■■■ MockInterview/ ← Q&A; interview interface
  ■ ■ ■■■ UI/ ← Shared: Button, Card, Badge, Modal
  ■ ■ ■■■ pages/
  ■ ■ ■■■ Landing.jsx
  ■ ■ ■■■ Dashboard.jsx
  ■ ■ ■■■ Analyze.jsx
  ■ ■ ■■■ JobAnalyzer.jsx
  ■ ■ ■■■ GapReport.jsx
  ■ ■ ■■■ Roadmap.jsx
  ■ ■ ■■■ Challenges.jsx
  ■ ■ ■■■ MentorChat.jsx
  ■ ■ ■■■ ResumeGenerator.jsx
  ■ ■ ■■■ PublicProfile.jsx
  ■ ■ ■■■ MockInterview.jsx
  ■ ■ ■■■ Settings.jsx
  ■ ■ ■■■ Leaderboard.jsx
  ■ ■ ■■■ NotFound.jsx
  ■ ■ ■■■ store/
  ■ ■ ■■■ useSkillForgeStore.js ← Zustand global state
  ■ ■ ■■■ hooks/
  ■ ■ ■■■ useGitHub.js
  ■ ■ ■■■ useRoadmap.js
  ■ ■ ■■■ useChallenge.js
  ■ ■ ■■■ utils/
  ■ ■ ■■■ api.js ← Axios instance + interceptors
  ■ ■ ■■■ helpers.js
  ■ ■ ■■■ App.jsx
  ■ ■ ■■■ main.jsx
  ■ ■■■ package.json
  ■ ■■■ tailwind.config.js
  ■
  ■■■ server/ ← Node.js Backend (GLM builds this)
  ■ ■■■ models/
  ■ ■ ■■■ User.js
  ■ ■ ■■■ Roadmap.js
  ■ ■ ■■■ Challenge.js
  ■ ■ ■■■ SkillGap.js
  ■ ■ ■■■ ChatHistory.js
```

```
■ ■■■ routes/
■ ■ ■■ auth.js ← GitHub OAuth routes
■ ■ ■■ github.js ← GitHub API proxy routes
■ ■ ■■ ai.js ← Gemini AI routes
■ ■ ■■ roadmap.js ← Roadmap CRUD
■ ■ ■■ challenges.js ← Daily challenge routes
■ ■ ■■ resume.js ← Resume bullet generator
■ ■ ■■ profile.js ← Public profile routes
■ ■■■ middleware/
■ ■ ■■ auth.js ← JWT verification
■ ■ ■■ cors.js
■ ■■■ services/
■ ■ ■■ githubService.js ← GitHub API calls
■ ■ ■■ aiService.js ← Gemini AI integration
■ ■ ■■ challengeScheduler.js ← node-cron daily job
■ ■■■ config/
■ ■ ■■ passport.js ← GitHub OAuth strategy
■ ■■■ index.js
■ ■■■ package.json
■
■ ■■ .env.example
■ ■■ .gitignore
■ ■■ README.md
```

08 — Database Design (MongoDB Schemas)

User Model — models/User.js

```
{ _id, githubId, username, displayName, avatarUrl, email,
  profileUrl, bio, publicRepos, followers, following,
  location, company, apiKey, aiProvider,
  theme: { type: String, default: 'dark' },
  isPublic: { type: Boolean, default: true },
  createdAt, updatedAt }
```

SkillGap Model — models/SkillGap.js

```
{ _id, userId (ref: User), jobTitle, jobDescription,
  matchScore: Number,
  userSkills: [{ skill, level, confidence }],
  jobRequirements: [{ skill, level, priority }],
  gaps: [{ skill, currentLevel, requiredLevel, gapSize, priority }],
  analyzedAt: Date }
```

Roadmap Model — models/Roadmap.js

```
{ _id, userId (ref: User), skillGapId (ref: SkillGap),
  duration: { type: Number, enum: [30, 60, 90] },
  weeks: [{ weekNumber, goal,
  days: [{ day, task, type, resource, completed, completedAt }]}],
  progress: Number, isActive: Boolean, createdAt }
```

Challenge Model — models/Challenge.js

```
{ _id, userId (ref: User),
  challenge: { title, description, type, difficulty, skill },
  date: Date, completed: Boolean, completedAt: Date,
  streak: Number, solution: String }
```

ChatHistory Model — models/ChatHistory.js

```
{ _id, userId (ref: User),
  messages: [{ role: 'user'|'assistant', content, timestamp }],
  sessionId: String, createdAt }
```

09 — GitHub OAuth Flow

SkillForge uses GitHub OAuth 2.0 for authentication. This means users log in with their GitHub account — no password required. Passport.js handles the strategy on the backend.

1. User clicks 'Login with GitHub' on Landing Page
2. Frontend redirects to: GET /api/auth/github
3. Backend redirects to GitHub OAuth authorization URL
4. User sees GitHub permission screen -> clicks 'Authorize SkillForge'
5. GitHub redirects back to: GET /api/auth/github/callback?code=xxx
6. Passport.js exchanges code for access token with GitHub
7. Backend fetches user profile from GitHub API using access token
8. User found in MongoDB? -> Update profile. New user? -> Create record.
9. Backend generates JWT token and sends it back to frontend
10. Frontend stores JWT in localStorage -> Redirects to /dashboard
11. All subsequent API calls include JWT in Authorization header
12. Backend middleware verifies JWT on every protected route

Required Environment Variables:

ENV Variable	Where To Get It	Example
GITHUB_CLIENT_ID	github.com/settings/developers -> New OAuth App	OWB3ApXXXXXXXXXX
GITHUB_CLIENT_SECRET	github.com/settings/developers -> New OAuth App	Abcd123ef456...
GITHUB_CALLBACK_URL	Set in OAuth App settings	http://localhost:5000/api/auth/github/callback
JWT_SECRET	Generate a random 32+ char string	your_super_secret_key_here
MONGODB_URI	MongoDB Atlas -> Connect -> Driver	mongodb+srv://user:pass@cluster...
GEMINI_API_KEY	aistudio.google.com/apikey	AlzaSyXXXXXXXXXX
CLIENT_URL	Your frontend URL	http://localhost:3000

10 — AI Integration Layer

All AI calls go through the Express backend as a secure proxy. The user's API key is sent in request headers — never stored on the server. Each AI feature has its own dedicated endpoint with a specific system prompt.

API Route	Feature	Prompt Strategy
POST /api/ai/analyze-gap	Skill Gap Analysis	System: expert career coach. Compare GitHub skills vs job requirements. Return JSON.
POST /api/ai/generate-roadmap	Roadmap Generator	System: learning architect. Given gaps and duration, generate week-by-week plan with
POST /api/ai/generate-challenge	Daily Challenge	System: coding coach. Generate one calibrated challenge for skill: {skill}. Return title, d
POST /api/ai/mentor-chat	AI Mentor Chat	System: senior developer mentor. Context: user stats, gaps, roadmap. Answer helpfully
POST /api/ai/resume-bullets	Resume Generator	System: tech recruiter. Turn repo data into 3 strong resume bullet points. Action verb +
POST /api/ai/mock-interview	Mock Interview	System: senior interviewer at {company}. Ask role-specific technical questions one at a
POST /api/ai/analyze-job	Job Listing Parser	System: JD analyst. Extract: required skills, experience level, tech stack, keywords. Re

11 — Deployment Strategy

Platform	Hosts	Cost	URL Pattern
Vercel	React frontend (client/)	Free	skillforge.vercel.app
Railway / Render	Node.js + Express (server/)	Free tier	skillforge-api.up.railway.app
MongoDB Atlas	Database (M0 free cluster)	Free forever	cloud.mongodb.com
GitHub	Source code + version control	Free	github.com/Kairo/skillforge

Deployment Steps:

1. Push code to GitHub main branch
2. Vercel detects push -> auto-builds and deploys frontend
3. Railway/Render detects push -> auto-deploys backend
4. Set environment variables in Vercel dashboard (VITE_API_URL)
5. Set environment variables in Railway (all server .env vars)
6. Update GITHUB_CALLBACK_URL to production URL in GitHub OAuth App
7. Test full OAuth flow on production -> Done!

12 — Development Roadmap (6 Phases)

Phase 1 — Project Setup & Auth

- Initialize client/ with React + Vite + Tailwind CSS
- Initialize server/ with Node.js + Express + Mongoose
- Connect MongoDB Atlas (create free cluster)
- Set up GitHub OAuth with Passport.js
- Build Landing Page with 'Login with GitHub' button
- Implement JWT generation and protected route middleware
- Deploy skeleton to Vercel (frontend) + Railway (backend)
- Test full OAuth login flow end-to-end

Phase 2 — GitHub Analyzer

- GitHub API service: fetch repos, languages, commits, stars
- Calculate language percentages and skill categories
- Build GitHub Analyzer page with repo cards
- Implement skill radar chart using Recharts
- Build GitHub contribution heatmap component
- Store analyzed data in User model

Phase 3 — AI Core Features

- Build Job Listing Analyzer page + AI job parser endpoint
- Implement Skill Gap Analysis AI endpoint + radar comparison UI
- Build Roadmap Generator with AI + timeline UI with checkboxes
- Implement Daily Challenge system with node-cron scheduler
- Build AI Mentor Chat with streaming + context injection
- Build Resume Bullet Generator page + export options

Phase 4 — Dashboard & Progress

- Build main Dashboard with stats cards + daily challenge card
- Implement progress tracking: roadmap %, streak counter
- Build Skill Progress Timeline component
- Build Mock Interview mode with AI Q&A; + evaluation
- Implement notification system for daily challenges

Phase 5 — Public Profile & Community

- Build Public Profile page (/u/:username)

- Implement privacy toggles per section
- Build Leaderboard page with streak rankings
- Add social share meta tags (og:image) for profile
- Build Settings page: API key, theme, privacy controls

Phase 6 — Polish & Launch

- Dark/Light mode full implementation
- Mobile responsive design audit of all pages
- Loading states, error boundaries, empty states
- PDF export for Roadmap and Skill Gap Report
- README.md + setup documentation
- Demo video recording for portfolio
- Add to portfolio: syed-humaam-ahmed.vercel.app

13 — Kimi 2.5/2.6 — Frontend Prompts

Paste these prompts into Kimi one at a time, in order. Kimi will generate all React frontend code.

Kimi Prompt 1 — Project Setup

Create a React + Vite + Tailwind CSS v3 project called SkillForge. Install dependencies: react-router-dom v6, zustand, axios, recharts, framer-motion, lucide-react. Set up the folder structure: src/components/, src/pages/, src/store/, src/hooks/, src/utils/. Create tailwind.config.js with dark mode support. Create a Zustand store in useSkillForgeStore.js that manages: user (null), isAuthenticated (false), theme ('dark'), apiKey (''), aiProvider ('gemini'). Create an Axios instance in src/utils/api.js that reads VITE_API_URL from .env and attaches JWT from localStorage to every request header.

Kimi Prompt 2 — Landing Page

Build the Landing Page (src/pages/Landing.jsx) for SkillForge. Dark theme (#0A0F1C background, #00D1FF accent, #7C3AED purple). Sections: (1) Hero: Big headline 'Bridge the Gap Between You and Your Dream Job', subtext, 'Login with GitHub' button that calls /api/auth/github. (2) Features section: 6 feature cards with icons (GitHub Analyzer, Skill Gap Report, AI Roadmap, Daily Challenges, AI Mentor, Resume Generator). (3) How It Works: 3 steps with icons. (4) CTA section at bottom. Use Framer Motion for smooth scroll animations.

Kimi Prompt 3 — Dashboard Page

Build the Dashboard page (src/pages/Dashboard.jsx). Layout: top Navbar + left Sidebar icon nav + main content area. Dashboard content: Welcome card showing user avatar and GitHub username. Stats row: 4 cards (Match Score, Roadmap Progress %, Streak Days, Challenges Done). Daily Challenge card with today's challenge + 'Start Challenge' button. Quick Actions: 4 buttons (Analyze GitHub, New Job Scan, Open Roadmap, Chat with Mentor). Recent Activity timeline. Use Recharts for the stats cards with animated number counters.

Kimi Prompt 4 — GitHub Analyzer Page

Build the GitHub Analyzer page (src/pages/Analyze.jsx). Calls GET /api/github/analyze on load. Shows: (1) Profile header: avatar, name, bio, followers, public repos. (2) Language radar chart using Recharts RadarChart — 8 axes: JavaScript, Python, Backend, Frontend, DevOps, Database, Mobile, AI/ML. (3) Repository grid: cards showing repo name, language, stars, last commit, description. (4) Contribution heatmap (grid of colored squares by week). (5) 'View Skill Gap' CTA button. Add loading skeleton states.

Kimi Prompt 5 — Skill Gap Report + Roadmap

Build two pages: (1) GapReport.jsx — Job description textarea input + 'Analyze Gap' button. Shows two RadarCharts side by side: 'Your Skills' vs 'Job Requirements'. Below: gap cards sorted by priority (red=critical, amber=moderate, green=strong). Match score big number with circular progress indicator. Export PDF button. (2) Roadmap.jsx — Active 30/60/90 day roadmap as a week-by-week accordion. Each day shows task + type badge + resource link + checkbox. Overall progress bar at top. 'Generate New Roadmap' button. Mark complete updates progress in real time.

Kimi Prompt 6 — AI Mentor Chat + Challenges + Resume

Build three pages: (1) MentorChat.jsx — Full-screen chat UI with streaming text support. Message bubbles (user right, AI left with avatar). Pre-built prompt chips at top: 'Explain my biggest gap', 'What should I build?', 'Start mock interview'. Input box + send. (2) Challenges.jsx — Today's challenge card (title, description, difficulty badge, skill tag). Streak tracker with flame icon. Submit/Complete button. History table of past challenges. (3) ResumeGenerator.jsx — Repo checkboxes to select projects. Generate button. Bullet point cards per project with edit, regenerate, copy icons. Export all button.

Kimi Prompt 7 — Public Profile + Settings

Build two pages: (1) PublicProfile.jsx (route: /u/:username) — Recruiter-friendly view. Profile header with avatar, name, bio. Stats row: match score, streak, challenges. Skills radar chart. Active roadmap summary. Privacy toggle banners. Copy profile URL button. (2) Settings.jsx — Sections: API Key Manager (provider dropdown + key input + save), Theme toggle (dark/light with preview), Privacy Controls (toggle public/private per section), Notification preferences toggle, Danger Zone (delete account button with confirmation modal). Use clean card-based layout with section dividers.

14 — GLM 5.1 — Backend Prompts

Paste these prompts into GLM 5.1 one at a time, in order. GLM will generate all Node.js backend code.

GLM Prompt 1 — Server Setup + Auth

Create a Node.js + Express.js backend for SkillForge. Install: express, mongoose, cors, dotenv, passport, passport-github2, express-session, connect-mongo, jsonwebtoken, bcryptjs, helmet. Create server/index.js that initializes Express, connects to MongoDB, applies CORS (allowing CLIENT_URL), sets up sessions, and mounts all routes. Create the User mongoose model with fields: githubId, username, displayName, avatarUrl, email, profileUrl, bio, publicRepos, followers, following, location, apiKey, aiProvider, theme, isPublic. Create server/config/passport.js with GitHub OAuth strategy using passport-github2. Create server/routes/auth.js with GET /api/auth/github, /api/auth/github/callback, /api/auth/me, /api/auth/logout. Callback should generate JWT and redirect to frontend.

GLM Prompt 2 — GitHub API Service

Create server/services/githubService.js that uses the GitHub REST API to: (1) fetchUserProfile(username, accessToken) — GET /users/:username. (2) fetchUserRepos(username, accessToken) — GET /users/:username/repos?per_page=100. (3) analyzeLanguages(repos) — aggregate language bytes across all repos, calculate percentages, map to skill categories: JavaScript->Frontend, Python->AI/ML, etc. (4) getContributionData(username) — last 52 weeks of commit activity. Create server/routes/github.js with: GET /api/github/analyze (fetch + analyze user's GitHub profile, store results in User model, return structured skills data). Add JWT auth middleware to all routes.

GLM Prompt 3 — AI Service + Skill Gap Route

Create server/services/aiService.js that integrates with Gemini API (using @google/genai). The API key comes from request headers (x-api-key) — never hardcoded. Functions: (1) analyzeSkillGap(userSkills, jobDescription, apiKey) — system prompt: expert career coach. Returns JSON: {matchScore, gaps:[{skill,currentLevel,requiredLevel,priority}]}. (2) generateRoadmap(gaps, duration, apiKey) — returns weeks array with daily tasks. (3) generateChallenge(skill, level, apiKey) — returns one coding challenge object. (4) generateResumeBullets(repoData, apiKey) — returns 3 bullet points per repo. (5) mentorChat(messages, userContext, apiKey) — streaming chat response. Create server/routes/ai.js mounting all AI endpoints.

GLM Prompt 4 — Roadmap + Challenge Routes

Create two route files: (1) server/routes/roadmap.js — POST /api/roadmap/generate (call aiService.generateRoadmap, save to Roadmap model), GET /api/roadmap/active (get user's active roadmap), PATCH /api/roadmap/complete-task (mark a day's task complete, update progress %), GET /api/roadmap/:id (get specific roadmap). (2) server/routes/challenges.js — GET /api/challenges/today (get today's challenge for user, create if not exists), POST /api/challenges/complete (mark complete, update streak), GET /api/challenges/history (paginated challenge history), GET /api/challenges/streak (current streak count). Create server/services/challengeScheduler.js using node-cron that runs at midnight to generate fresh challenges for all active users.

GLM Prompt 5 — Resume + Profile + Settings Routes

Create three route files: (1) `server/routes/resume.js` — `POST /api/resume/generate` (takes selected repo IDs, calls `aiService.generateResumeBullets`, returns bullets array), `POST /api/resume/export-pdf` (generate simple text PDF of all bullets). (2) `server/routes/profile.js` — `GET /api/profile/:username` (public profile — no auth required, return only public sections), `PATCH /api/profile/privacy` (toggle section visibility), `GET /api/profile/leaderboard` (top 20 users by streak). (3) `server/routes/settings.js` — `PATCH /api/settings/api-key` (save encrypted API key), `PATCH /api/settings/theme` (save theme preference), `DELETE /api/settings/account` (delete all user data). All protected routes require JWT middleware.

15 — Google Antigravity — Fallback Prompts

If Kimi + GLM don't produce satisfactory results, paste these into Google Antigravity in order. Antigravity will build the full project.

Antigravity Prompt 1 — Full Project Overview

Build a full-stack web application called SkillForge — an AI-powered developer growth platform. Frontend: React + Vite + Tailwind CSS v3. Backend: Node.js + Express.js. Database: MongoDB with Mongoose. Auth: GitHub OAuth 2.0 with Passport.js + JWT. AI: Google Gemini API (user provides own key via BYOK model). Color theme: #0A0F1C background, #00D1FF cyan accent, #7C3AED purple, #10B981 green. The app lets developers: log in with GitHub, analyze their GitHub profile for skills, paste job listings to get AI skill gap analysis, receive personalized 30/60/90 day roadmaps, get daily coding challenges, chat with an AI mentor, generate resume bullet points from GitHub activity, and share a public recruiter-friendly profile page. Set up the complete monorepo folder structure with client/ and server/.

Antigravity Prompt 2 — Auth + Dashboard

Implement GitHub OAuth 2.0 authentication for SkillForge. Backend: Passport.js with passport-github2 strategy. Routes: /api/auth/github, /api/auth/github/callback, /api/auth/me, /api/auth/logout. On successful auth, generate JWT, store in localStorage on frontend, redirect to /dashboard. Build the Landing page with hero section, features grid, and 'Login with GitHub' button. Build the Dashboard with: welcome card showing GitHub avatar and username, 4 stats cards (Match Score, Roadmap Progress, Streak, Challenges Done), Daily Challenge card, Quick Actions grid, and Recent Activity.

Antigravity Prompt 3 — GitHub Analyzer + Skill Gap

Build the GitHub Analyzer feature: fetch user's repos and languages from GitHub API, calculate skill percentages, display as a radar chart (Recharts RadarChart) across 8 categories, show repo cards grid, contribution heatmap. Build the Skill Gap feature: job description input textarea, call Gemini AI to extract requirements, compare against user skills, display two side-by-side radar charts (Your Skills vs Job Requirements), show color-coded gap cards, calculate and display a Match Score percentage with circular progress indicator.

Antigravity Prompt 4 — Roadmap + Challenges + AI Chat

Build three features: (1) Roadmap Generator: AI generates 30/60/90 day learning plan from skill gaps. UI: week-by-week accordion with daily tasks, checkboxes, resource links, overall progress bar. (2) Daily Challenges: node-cron generates fresh challenges at midnight per user's skill gaps. UI: challenge card with difficulty badge, streak flame tracker, history table. (3) AI Mentor Chat: streaming Gemini chat with full user context injected. UI: full-screen chat with message bubbles, pre-built prompt chips, streaming text.

Antigravity Prompt 5 — Resume + Public Profile + Deploy

Build two final features: (1) Resume Bullet Generator: user selects repos, AI generates 3 professional bullets per repo (action verb + tech + outcome), user can edit/copy/export. (2) Public Profile page (/u/:username): recruiter-friendly view with stats, radar chart, roadmap summary, privacy toggles, shareable URL copy button. Also build: Settings page (API key manager, theme toggle, privacy controls), Leaderboard page (top streaks), 404 page. Configure Vercel deployment for frontend and Railway for backend. Write complete README.md with setup instructions.

Let's Build It.

Try Kimi first for frontend • GLM for backend • Antigravity if needed

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